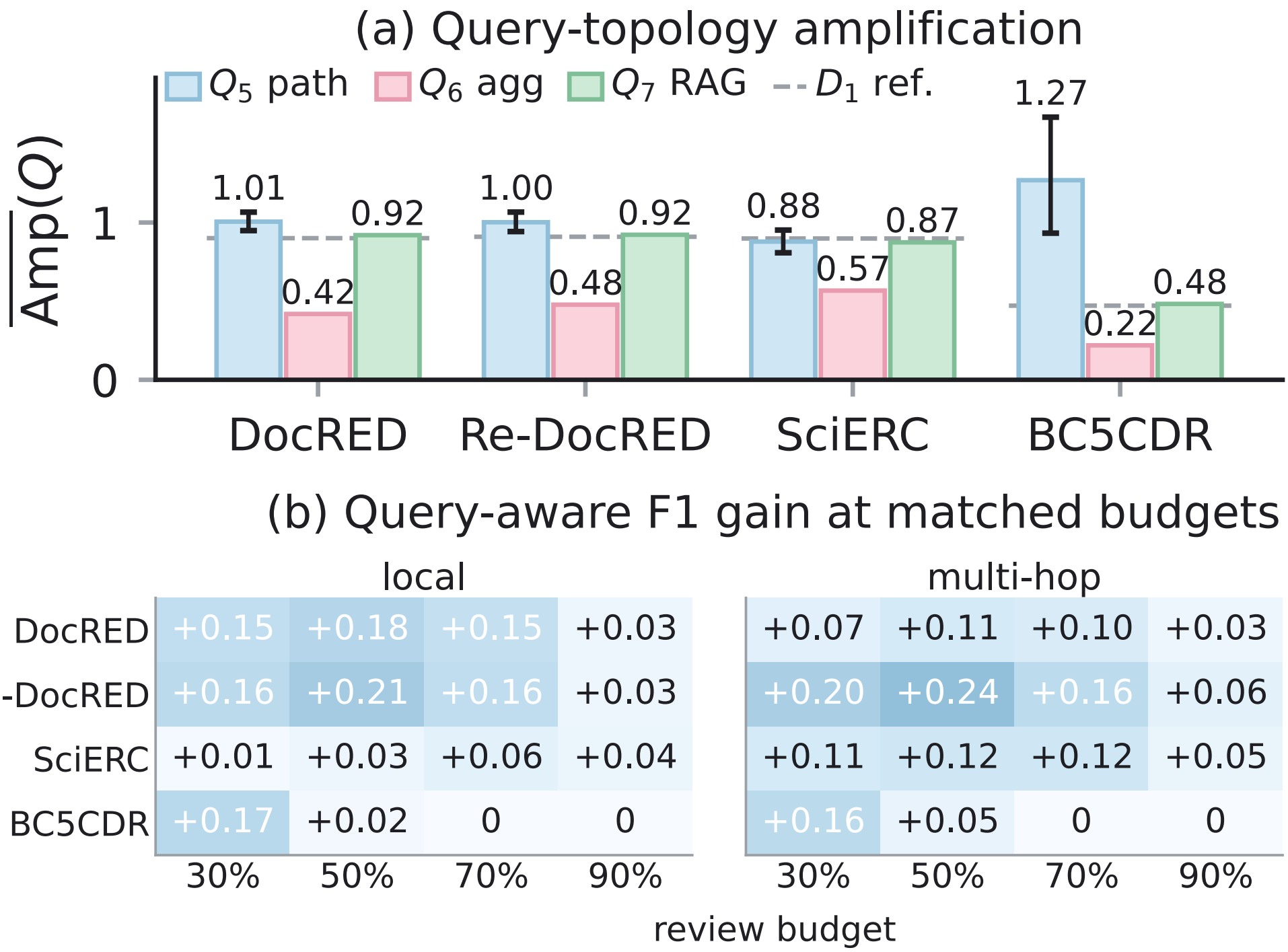
(a) Query-topology amplification
$Q_5$ path
$Q_6$ agg
$Q_7$ RAG
$D_1$ ref.
$\overline{\mathrm{Amp}(Q)}$
1
0
1.01
0.42
0.92
1.00
0.48
0.92
0.88
0.57
0.87
1.27
0.22
0.48
DocRED
Re-DocRED
SciERC
BC5CDR
(b) Query-aware F1 gain at matched budgets
local
multi-hop
DocRED +0.15 +0.18 +0.15 +0.03 +0.07 +0.11 +0.10 +0.03
Re-DocRED +0.16 +0.21 +0.16 +0.03 +0.20 +0.24 +0.16 +0.06
SciERC +0.01 +0.03 +0.06 +0.04 +0.11 +0.12 +0.12 +0.05
BC5CDR +0.17 +0.02 0 0 +0.16 +0.05 0 0
30% 50% 70% 90%
30% 50% 70% 90%
review budget